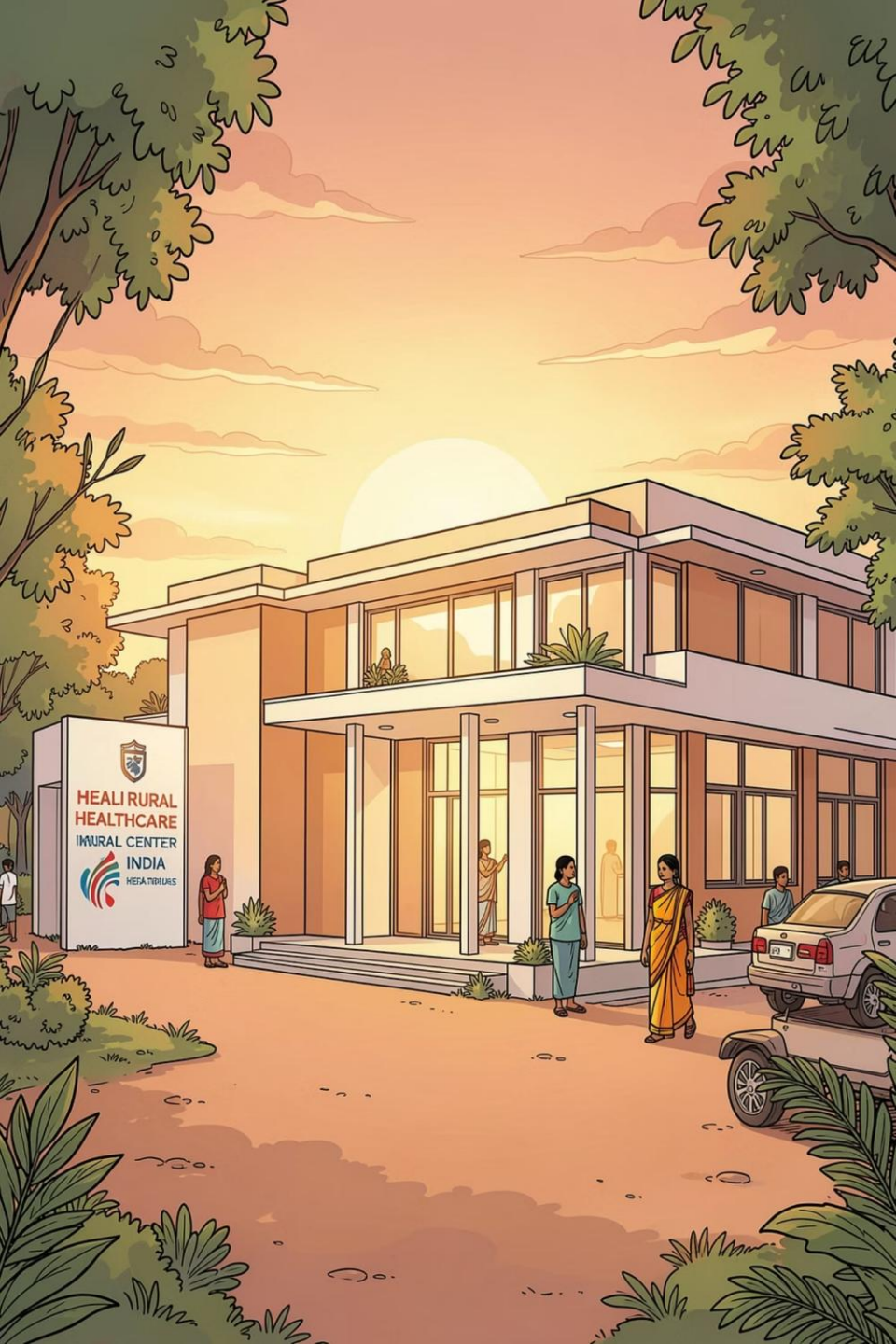




UNIFIED HEALTHCARE OPERATING SYSTEM

UHOS

Unified Healthcare Operating System

An explainable, event-driven operations platform for India's PHCs and CHCs. UHOS replaces fragmented paper registers and phone-based coordination with a single, auditable event log — giving district health officers, doctors, and citizens one shared source of truth.

The Problem

Primary health infrastructure runs on paper, phone calls, and guesswork

24%

Stockouts unnoticed

of PHC medicine stockouts go unnoticed until a patient is turned away

0

Real-time visibility

district officers have into facility-level stock or staffing today

3

Days of lag

typical delay between a shortage occurring and anyone acting on it

District officers currently rely on end-of-month paper reports to discover which facility ran out of medicine, which doctor didn't show up, or which ward is over capacity. By the time anyone finds out, the patient has already been turned away — and the trust deficit grows.

Existing Systems

Why current solutions fall short

Four categories of tools dominate the landscape today. Each addresses a piece of the problem, but none fit the operational reality of a two-doctor PHC or the oversight needs of a district health officer.



Manual registers

Paper-based, no cross-facility visibility, reconciled monthly at best — by then the data is history, not intelligence.



Generic hospital ERPs

Built for large multi-specialty hospitals. Too heavy, too expensive, and too complex for a two-doctor PHC to adopt.



Black-box ML dashboards

Predictions with no explanation. A health worker can't act on a number they don't trust — and shouldn't have to.

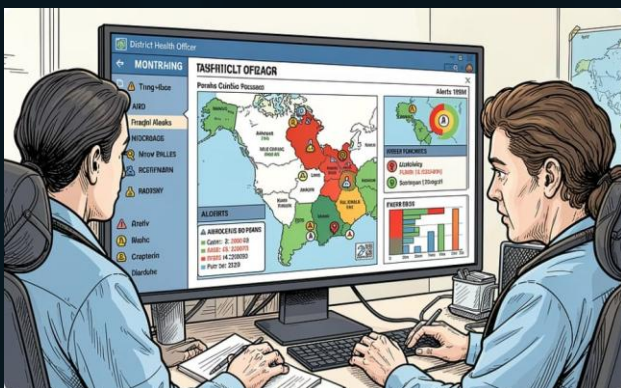


Siloed point solutions

Separate tools for inventory, staffing, and appointments that never talk to each other — data drifts, context is lost.

Our Solution

One event-driven backbone, three purpose-built apps — every decision explained



District Dashboard

Live command center for district officers — facility scores, alerts, and a real-time event timeline showing everything happening across every PHC and CHC in the district as it happens.



Doctor Workspace

A prescribing flow that shows the AI's exact reasoning behind every stock decision, surfaced through a transparent Why drawer. No black boxes — just auditable logic the doctor can verify and trust.



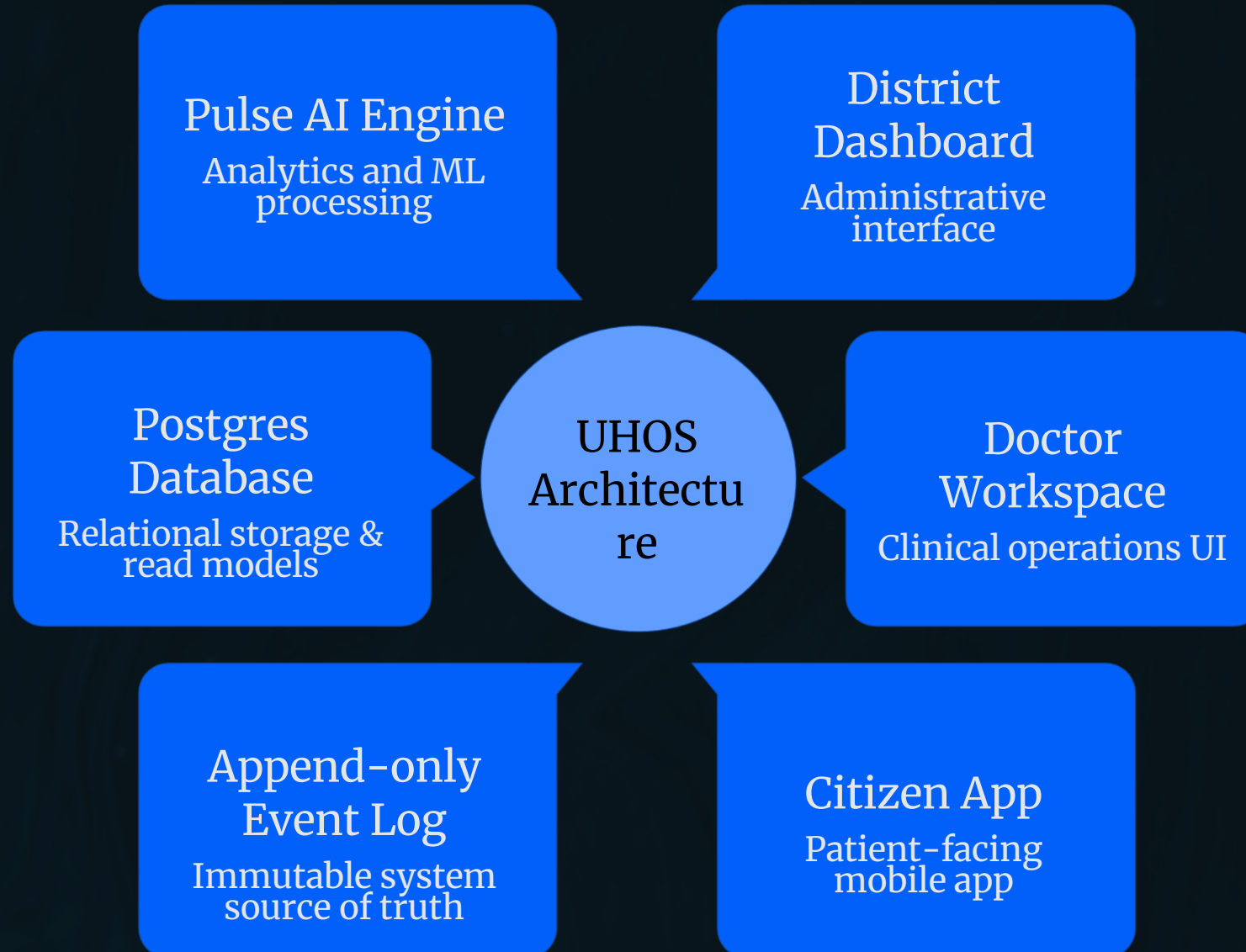
Citizen App

Doctor availability and facility information in the citizen's own language — English, Hindi, Kannada, or Gujarati. No login required to browse; designed for the person walking into a PHC, not the administrator running it.

- ❑ All three apps run on a single append-only event log — nothing is duplicated, nothing drifts out of sync.

System architecture

UHOS is built on an append-only event log at its core. Every action — a prescription written, a stock check-in, a doctor marking attendance — is recorded as an immutable event. Three independent applications read from and write to this shared backbone, ensuring perfect consistency without the overhead of a message broker.



Pulse AI

Explainable by design

Pulse AI doesn't use black-box machine learning. Instead, it applies plain, auditable formulas over real inventory and consumption data — so every recommendation can be traced, understood, and challenged by the health worker on the ground.

Transparent recommendations

Every output shows the exact stock level, consumption rate, and threshold it used — displayed inline, never hidden behind a menu.

Smart redistribution

Surplus transfers suggest the nearest PHC with available stock — never a mystery transfer to an unknown facility.

Auditable facility scores

Facility scores combine medicine and doctor availability into one number reproducible from the raw event log at any time.

Decision logging

Every AI decision is written to a dedicated decision log — available for judicial or administrative review, with full reproducibility.

From prescription to citizen app

The end-to-end journey shows how a single event ripples through the system. A doctor prescribes medicine in the Doctor Workspace; Pulse AI checks stock and suggests redistribution if needed; the event log updates instantly; the District Dashboard reflects the change; and the Citizen App shows updated availability — all from one canonical event.



Doctor Prescribes

Pulse AI Checks

Event Logged

Dashboard Updates

What used to take a month of reconciliation now propagates in milliseconds — and every stakeholder sees the same truth.

Operations

Healthcare operations beyond medicine

The operational signals district officers actually ask for — not just stock levels, but the full picture of whether a facility is functioning. Every metric is derived live from the event log, so it's never stale and never requires a manual refresh.



Attendance

Doctor check-ins feed directly into facility scoring — no separate attendance register, no end-of-month reconciliation gap.



Beds

Available capacity always derived live from the event log — never stored, never stale, always reflecting the current moment.



Footfall

Patient volume trends per facility over time — spotting surges early so resources can be redirected before a crisis.



Test availability

Which diagnostic tests are actually offered, facility by facility — so referrals go to centres that can actually deliver.

Citizen App

Built for the person walking into a PHC

Most healthcare platforms stop at the administrator. UHOS extends to the citizen: a lightweight, multilingual app that requires no login to browse. Doctor availability, facility information, and medicine stock are all surfaced in a clean, accessible interface.

Doctor availability

Real-time information on which doctors are present, updated as they check in — no more wasted trips to a closed facility.

Facility information

Services offered, operating hours, test availability — all maintained from the same event log that powers the district dashboard.

Multilingual: EN / HI / KN / GU

Language switch is a first-class feature, not a translated afterthought. Every string ships in all four languages at launch, verified by an automated smoke test suite.

Technical Stack

Built with a modern, lightweight, fully open-source stack

UHOS is designed for rapid deployment and zero-friction local development — from a district pilot to a national rollout.

Backend

FastAPI + SQLAlchemy. Postgres in production, SQLite for zero-setup local development.

Frontend

Next.js 15, React 19, TypeScript, and Tailwind CSS for a responsive, accessible interface.

Real-time

Native WebSocket tailing the append-only events table — no message broker, no infrastructure overhead.

Testing

40 backend pytest cases + 21 frontend vitest cases, plus a full multilingual smoke test suite for every release.

Deployment

Dockerized end-to-end. Deploy to Railway/Render + Vercel, or run one-command docker compose locally.

Key statistics at a glance

A visual snapshot of the core UHOS metrics, arranged as a dashboard for faster scanning and stronger impact.

6

Districts

Community districts currently in scope for rollout and coordination.

61

Reports

Operational reports and insights tracked through the platform.

4

Core modules

Integrated modules supporting the main UHOS workflow.

3

Phases

Delivery phases planned to guide implementation and adoption.

Future scope

Our roadmap extends UHOS capabilities to address even broader healthcare challenges and opportunities, focusing on expanding functionality and integration.



Role-based audit

Strengthen accountability with role-based access and clear audit trails for every critical action.



SMS/IVRS fallback

Keep patients connected when digital access is limited by adding reliable SMS and IVRS backup channels.



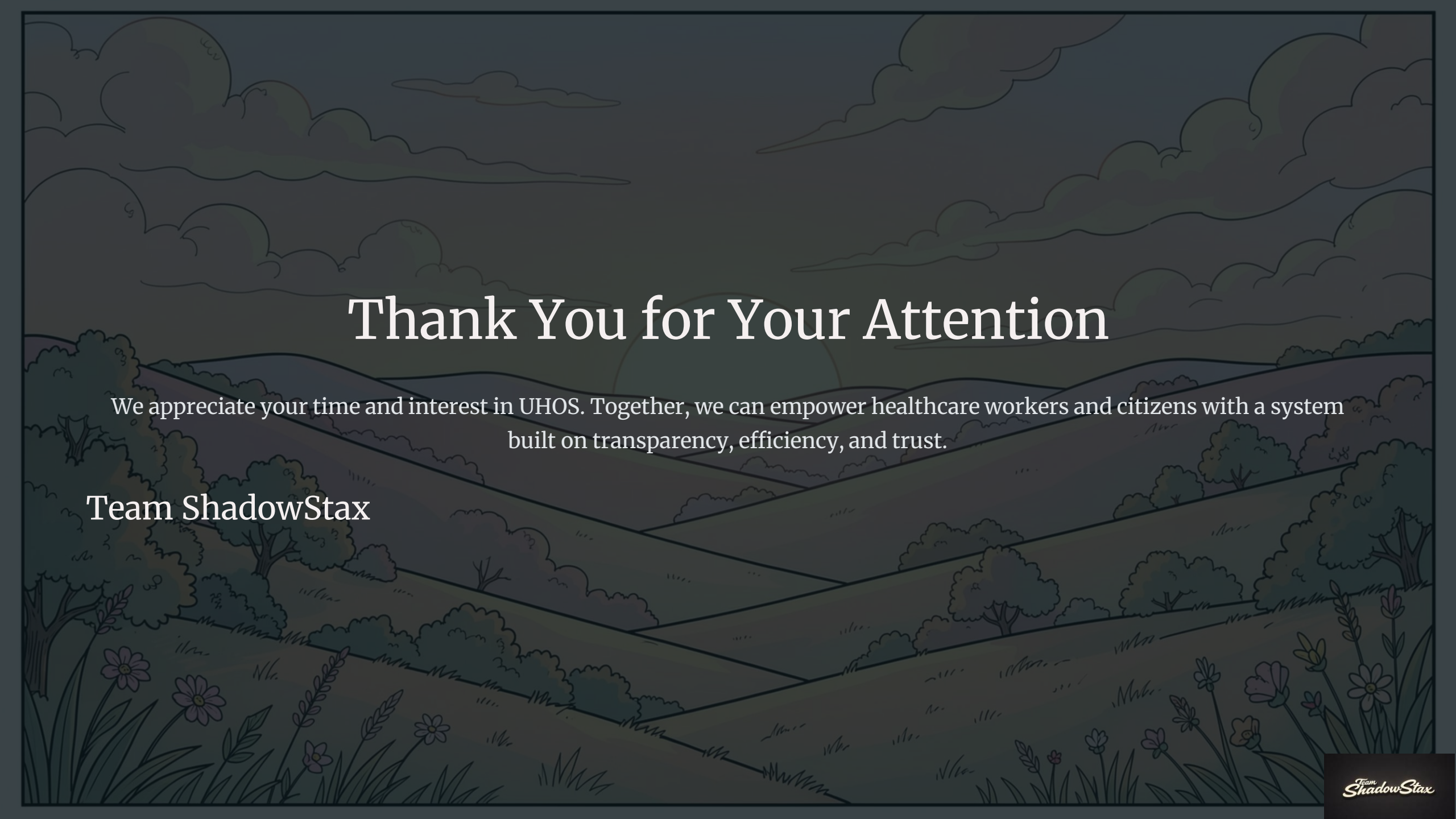
State health integration

Align UHOS with state health systems to improve data sharing, coordination, and reporting.



Pulse AI forecasting

Use Pulse AI to anticipate demand, surface trends, and support better planning decisions.



Thank You for Your Attention

We appreciate your time and interest in UHOS. Together, we can empower healthcare workers and citizens with a system built on transparency, efficiency, and trust.

Team ShadowStax